

Agentic Delegation and the Language Frontier

Quispe and Xu (2026): model, evidence, and an independent
Lean audit

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Artificial Intelligence and Economic Modeling

github.com/amchavez/ai-03-quispe

The citation described an earlier draft, not arXiv v2

Citation received

Quispe (2026)

Coding Beyond Your Training: Claude Code and the Technological Frontier of Software Developers

Verified v2

Alexander Quispe and Kevin Xu

Agentic Delegation and the Language Frontier of Software Developers: A Model and Evidence from Claude Code on GitHub

Revised July 7, 2026

What changed

The received title is acknowledged in the v2 source as an earlier version; authorship, title, framing, and evidentiary scope differ.

Source: Quispe and Xu, arXiv:2605.25438v2, title page and submission history.

The paper asks whether AI expands what a developer can produce

Question

Can agentic coding assistants make unfamiliar-language opportunities viable?

Contribution

Agentic AI adds a new production mode: the agent executes while the developer specifies, decomposes, and verifies.

Central distinction: observed production, knowledge, general productivity, and agentic delegation are different objects.

Source: Quispe and Xu, Sections 1 and 4; Remark “What the model does and does not claim.”

Frontier object

Languages in which the developer
ships work

≠ languages the developer has
learned to code unassisted

The developer follows a menu-and-threshold rule

$$V^S = \omega + s\mu - \frac{\rho s^2}{2\pi} - b,$$

$$V^C = V^S + \gamma s - r_C,$$

$$V^D = \omega + (1 - \lambda)s\mu + \lambda az(A) - \kappa - r_D \\ - \frac{\rho}{2} \left[\frac{(1 - \lambda)^2 s^2}{\pi} + \sigma_D^2 \right].$$

Primitives

- ▶ s : language-specific skill
- ▶ a : general specification ability
- ▶ ω : opportunity value
- ▶ b : activation cost
- ▶ ρ : risk aversion
- ▶ λ : delegated execution share

$$V^g = \max_{m \in \mathcal{M}_g} V^m, \quad Z^g = 1\{V^g \geq 0\}.$$

Source: Quispe and Xu, Section 4.1, Equations (1)-(3).

Only delegation moves unfamiliar-language entry

$$T^S = b - s\mu + \frac{\rho s^2}{2\pi},$$

$$T^1 = T^S - \max\{0, \gamma s - r_C\},$$

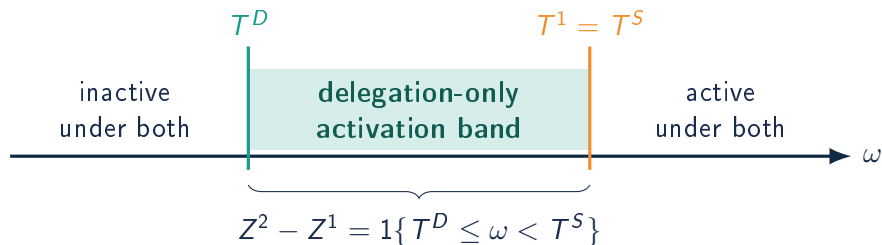
$$T^D = b - (1 - \lambda)s\mu - \lambda az(A) + \kappa + r_D + \frac{\rho}{2} \left[\frac{(1 - \lambda)^2 s^2}{\pi} + \sigma_D^2 \right].$$

Assumption 1 supplies the separation

For unfamiliar languages, $\gamma s - r_C \leq 0$, hence $T^1 = T^S$. Conversational augmentation does not move entry; delegation can if $B \equiv T^S - T^D > 0$.

Source: Quispe and Xu, Section 4.2, Equations (4)-(7), pp. 14-15.

Delegation activates a middle band of opportunities



Proposition 1: menu inclusion gives $Z^2 \geq Z^1$ path by path.

Proposition 2: with $B > 0$ and continuous F , activation probability is $F(T^S) - F(T^D) \geq 0$.

Source: Quispe and Xu, Section 4.2, Propositions 1-2, pp. 15-16.

The stock can keep growing after the first-use flow peaks

Proposition 3:

$$\Delta C_i(s) = \sum_{k \in \mathcal{U}_i} [(1 - p_{ik}^1)^{s+1} - (1 - p_{ik}^2)^{s+1}] \geq 0.$$

Flow

Newly used languages can spike at adoption and then fall as the at-risk set is depleted.

Stock

Cumulative languages retain first uses, so the gap can continue to rise.

Printed strict claim: under $p^1 = 0 < p^2$, the paper states strict increase and concavity over the observed horizon. The allowed endpoint $p^2 = 1$ breaks both strict inequalities.

Source: Quispe and Xu, Section 4.3, Proposition 3, p. 17; Theory Appendix proof.

At adoption, language breadth rises sharply

Outcome	$ATT(0)$	SE
Active languages	+2.528	0.063
Newly used languages	+1.193	0.051
Language entropy	+0.382	0.009
Cumulative languages	+1.604	0.054

5,346 developers; 28 months; 3.15 million commits; 57.2 million changed files.

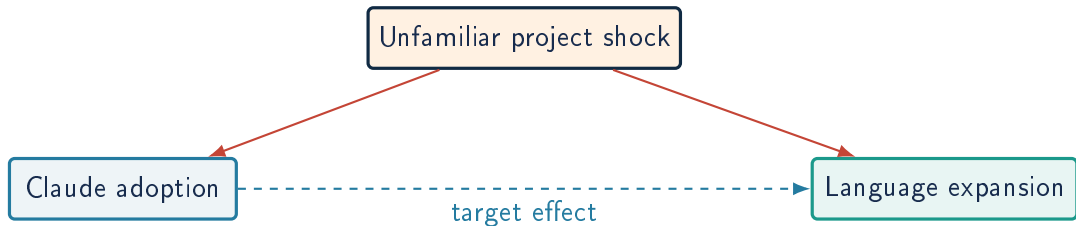
Source: Quispe and Xu, Sections 5 and 7; Table 2; source table `table_cl_main.tex`.

Economic magnitude

Pre-adoption active-language mean: **0.90**. The active set roughly triples on impact.

The cumulative series has non-trivial pre-trends and is treated as descriptive.

The event study does not identify causality



- ▶ Staggered DiD handles cohort heterogeneity and avoids already-treated comparisons.
- ▶ Removing Claude commits and the first-Claude language addresses mechanical measurement.
- ▶ Neither resolves project-driven adoption; activity already rises at $e = -1$.

Conclusion: robust event-time associations consistent with delegation, not a causal estimate.

Source: Quispe and Xu, Section 9, "Discussion: Identification Threats."

AI primitives imply expansion or deskilling

Quispe and Xu

AI adds specification and delegated execution:

$$\{S, C\} \longrightarrow \{S, C, D\}.$$

A lower threshold activates unfamiliar-language opportunities: the extensive margin expands.

Aouad, Lykouris, and Zhong

Skill, effort, and AI assistance are substitutable inputs:

$$x = s + e + a, \quad \max_{e \geq 0} p(x) - \gamma e.$$

More AI can reduce effort; with endogenous skill or unreliable assistance, productivity can fall and skills can polarize.

Modeling contrast: menu expansion creates a feasible margin; input substitution can cannibalize effort and skill formation.

Source: Quispe and Xu, Sections 1 and 4; Aouad, Lykouris, and Zhong (2026), Sections 1-5, arXiv:2605.11350v1.

Lean requires $p_2 < 1$ for strict dynamics

Printed: $p_1 = 0 < p_2 \leq 1 \Rightarrow G_s < G_{s+1}$ and $\Delta^2 G_s < 0$. **But at $p_2 = 1$:**
 $G_s = 1 - (1 - 1)^{s+1} = 1$.

Corrected Spec boundary

```
dynamicCumulativeLanguageEffectCorrectedSpec
0 <= p1 <= p2 <= 1 -> weakGap
p1 = 0 and 0 < p2 < 1
-> strictIncrease and strictConcavity
```

Formal refutation

```
theorem ...EndpointCounterexample :
  not ...PrintedStrictClaim := by
  intro h
  have hUnit := h Unit (fun _ => 1) ...
  have hFalse := hIncreasing 0
  norm_num at hFalse
```

Five endpoints compile with zero 'sorry', 'admit', or axioms; build and 'check -fast' succeed.

Verdict: partially formalized.

Source: lean/PaperInterface.lean; MainTheorems.lean; ProofInterface.lean; validation report.

The handwritten check makes saturation visible

Proposición 3, ecuación del cumulative gap :

$$G_s = (1-p_1)^{s+1} - (1-p_2)^{s+1}$$

$$p_1 = 0, \quad p_2 = 1$$

$$G_s = (1-0)^{s+1} - (1-1)^{s+1} \\ = 1^{s+1} - 0^{s+1} = 1$$

$$G_{s+1} - G_s = 1 - 1 = 0$$

$$(G_{s+2} - G_{s+1}) - (G_{s+1} - G_s) = 0 - 0 = 0$$

∴ Por lo tanto, no hay crecimiento ni concavidad estrictos; si $G_s \geq 0$

∴ El endpoint $p_2 = 1$ produce saturación inmediata. El resultado estricto requiere $0 < p_2 < 1$.

Two-minute derivation

Set one language, $p_1 = 0$,
 $p_2 = 1$. Then show:

$$G_s = (1-0)^{s+1} - (1-1)^{s+1} = 1.$$

Hence $G_{s+1} - G_s = 0$,
contradicting strict growth; the
second difference is also zero.

Source: Quispe and Xu, Proposition 3 and its Theory Appendix proof; independent Lean endpoint audit.

Expansion does not uniquely identify delegation

Extension: generic productivity

Replace skill-complementary augmentation with

$$V^P = V^S + q - r_P, \quad T^P = T^S - (q - r_P).$$

If $q > r_P$, even a non-agentic shock creates an activation band. Expansion alone does not reveal whether T^P or T^D moved.

Bottom line: a large shift in production breadth; five compiled Lean cores; one strict-endpoint defect; causal and mechanism-specific identification remain open.

Source: Quispe and Xu, Theory Appendix and Sections 8-10; extension developed in `extensions.md`.

Discriminating test

Cross post-adoption change with:

- ▶ pre-adoption portfolio breadth,
- ▶ independent verification ability,
- ▶ task-level delegation intensity.

Delegation predicts a positive interaction; a common productivity shift does not.